Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/07/2026
PI Name:	Bharti, Bhuvnesh Ph.D.		
Project Title:	Colloidal Assembly: Understanding the Electric Field Driven Assembly of Colloids and Its Applications		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baton Rouge	**State:**	LA
Zip Code:	70803	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR - Physical Sciences
Start Date:	08/01/2024	**End Date:**	07/31/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Colleen Fava is the Louisiana EPSCoR Director. Ali Shaykhian (ali.shaykhian@nasa.gov) is Technical Monitor, and Timothy P. Griffin (timothy.p.griffin@nasa.gov) is Technical Officer for this grant.		
COI Name (Institution):	Miller, Mary (Baton Rouge Community College) Pasika, Noshir (Tulane University) Peters, Andrew (Louisiana Tech University) Zhang, Donghui (Louisiana State University)		
Grant/Contract No.:	80NSSC24M0105		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The primary research objective of this project is to test a hypothesis that the phase behavior and interlinking of colloids in geometric confinement can be precisely controlled by programming the characteristics of the colloidal particles and external electric field induced interactions in microgravity environments. Based on the fundamental understanding gained in this project, the team will investigate a new concept for self-healing fissures by electrically assembling, jamming, and gelling the colloidal dispersions in the confined 'microcracks'. Previous studies have reported methods to synthesize self-repairing materials; however, such methods suffer from limitations, including the requirement for distinct mechanisms of crack detection and repair. The novelty of the proposed work is in using an electric field as the only stimulus to "self-activate" upon crack formation and rapidly drive the assembly of colloids within the crack leading to its repair. 1. Objective 1: Determine the influence of E field, particle shape, and microgravity on the directed assembly of colloids under confinement. 2. Objective 2: Measure the adhesion and friction forces between polymer grafted surfaces and determine the influence of applied fields on colloidal assembly. 3. Objective 3: Functionalize particles with thermally induced cross-linkable polymer grafts for 'on-demand' formation of mechanically stable colloidal aggregates. 4. Objective 4: Optimize and determine the kinetics of crack repair and its mechanical strength using a lab-based prototype for testing the self-healing capability of the colloidal dispersions. 5. Objective 5: Provide resources and opportunities to students at minority serving institutes (MSIs) by offering paid research internships in team's laboratories and supporting their visits to NASA Glenn Research Center (NASA-GRC).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The team aims to develop a fundamental understanding of the directed physicochemical assemblies of colloidal particles driven by an electric field within confinement on Earth and in microgravity.
Task Progress:	The project is developing new ways to understand and guide the movement and organization of tiny particles using electric fields, with the long-term goal of supporting advanced repair technologies for aerospace materials. During the first reporting period, the team established key experimental and modeling capabilities, built new collaborations across participating institutions, trained researchers in shared methods, and began preparing specially designed particle materials needed for future testing. Early work has shown that electric fields can direct particles into different arrangements under controlled conditions, providing an important foundation for later studies related to material repair. The team also began developing laboratory setups and protocols that will support future testing under conditions relevant to aerospace applications. The project has also strengthened research infrastructure by connecting expertise in materials, interfaces, modeling, fabrication, and biological testing across institutions. Researchers have exchanged training, shared experimental approaches, and coordinated plans for the next phase of the work. Results are still preliminary because the project began recently, and public dissemination will be handled carefully through presentations, publications, and NASA reporting, while protecting potentially patentable concepts. In the next reporting period, the team will expand testing of particle organization, prepare equipment for drop-tower studies, evaluate material bonding behavior under applied fields, and provide undergraduate research opportunities connected to the project. Note: Technical and scientific details have been intentionally omitted from this public summary because the work may lead to an invention disclosure. A complete scientific report containing the full technical details has been submitted to NASA and is not publicly accessible at the time of submission of this Task Book reprot.
Bibliography Type:	Description: (Last Updated:)